Architectural Drivers

ARCHITECTURALLY SIGNIFICANT REQUIREMENTS

Functional vs. Non-Functional Requirements

NOT CALLED NON-FUNCTIONAL ANYMORE

Technical vs. Business Constraints

Business: "We must launch by Black Friday," "Budget is $5k/month.“

Technical: "Must integrate with 20-year-old COBOL mainframe," "Team only knows Python."

The Three Pillars of a Request

1. Functional: What it does (User can login).
2. Non-Functional (Quality Attributes): How it behaves (Login takes < 2s).
3. Constraints: What limits us (Must use AWS).

The “-ilities” (Quality Attributes)

Availability: Is the system up?

Scalability: Can it handle more users?

Maintainability: Can we fix it easily?

Security: Is the data safe?

Performance vs. Scalability

Performance: A single user's experience (Latency).

Scalability: The system's capacity for 1,000,000 users (Throughput).

Quality Attributes

99.9%

Reliability

Percentage indicating the dependable uptime and correct functioning of systems.

10K+

Transactions

Signifies the scale of operations a system can handle, testing the limits of performance and throughput.

0.1s

Latency

Threshold for response times, indicating swift and efficient system reactions.

Extracting characteristics from requirements.

Some characteristics come from explicit statements in requirements documents.

For example, expected number of users and scale commonly appear explicitly.

Extracting characteristics from domain concerns.

An architect must be able to translate domain concerns into characteristics.

Is scalability the biggest concern? Or is it performance? Perhaps security?

When collaborating with domain stakeholders try to make the list as short as possible.

Each characteristic supported, complicates the design.

The number of characteristics is less important than the simplicity of the design.

Extracting characteristics from domain concerns.

A hint is to make the stakeholders select the top three characteristics.

This can lead to a discussion among all parties on what is important.

The issue comes with wording and terms translations: architects and stakeholders may have different terms for each concept.

For the most frequent concerns we can select a number of characteristics that may apply to them.

Extracting characteristics from domain concerns.

Domain Concern.	Architecture Characteristics.
Mergers and Acquisitions.	Interoperability, scalability, adaptability, extensibility.
Time to market.	Agility, testability, deployability.
User satisfaction,	Performance, availability, fault tolerance, testability, deployability, agility, security.
Competitive Advantage.	Agility, testability, deployability, scalability, availability, fault tolerance.
Time and budget.	Simplicity, feasibility.

Taxonomy

ISO 25010 Quick Reference Guide

Quality Characteristic	Brief Description
Functional Suitability	The degree to which a product provides functions that meet stated and implied needs, including functional completeness, correctness, and appropriateness.
Performance Efficiency	The degree to which a system performs within specified time and throughput parameters while remaining efficient in its use of resources like CPU and memory.
Compatibility	The ability of a product to exchange information with other systems or share a common environment and resources without detrimental impact (co-existence).
Interaction Capability	The degree to which a product can be used by specific users to complete tasks; this encompasses usability, learnability, and user engagement.
Reliability	The degree to which a system performs specified functions under stated conditions for a period of time, including its ability to recover from crashes.

ISO 25010 Quick Reference Guide

Quality Characteristic	Brief Description
Security	The degree to which a product protects information and data so that persons or other systems have the appropriate level of data access and authorization.
Maintainability	The effectiveness and efficiency with which a system can be modified by developers to fix bugs, improve performance, or add new features.
Flexibility	The degree of efficiency with which a product can be adapted for different hardware, software, or other operational environments (includes scalability and adaptability).
Safety	The degree to which a system limits the risk of harm to people, business, or the environment, focusing on operational constraints and fail-safe integration

Quality attributes: ISO 25010

SOFTWARE PRODUCT QUALITY								
FUNCTIONAL SUITABILITY	PERFORMANCE EFFICIENCY	COMPATIBILITY	INTERACTION CAPABILITY	RELIABILITY	SECURITY	MAINTAINABILITY	FLEXIBILITY	SAFETY
FUNCTIONAL COMPLETENESS FUNCTIONAL CORRECTNESS FUNCTIONAL APPROPRIATENESS iso25000.com	TIME BEHAVIOUR RESOURCE UTILIZATION CAPACITY	CO-EXISTENCE INTEROPERABILITY	APPROPRIATENESS RECOGNIZABILITY LEARNABILITY OPERABILITY USER ERROR PROTECTION USER ENGAGEMENT INCLUSIVITY USER ASSISTANCE SELF-DESCRIPTIVENESS	FAULTLESSNESS AVAILABILITY FAULT TOLERANCE RECOVERABILITY	CONFIDENTIALITY INTEGRITY NON-REPUDIATION ACCOUNTABILITY AUTHENTICITY RESISTANCE	MODULARITY REUSABILITY ANALYSABILITY MODIFIABILITY TESTABILITY	ADAPTABILITY SCALABILITY INSTALLABILITY REPLACEABILITY	OPERATIONAL CONSTRAINT RISK IDENTIFICATION FAIL SAFE HAZARD WARNING SAFE INTEGRATION

Operational Architectural characteristics

TERM	DEFINITION
Availability	How long the system needs to be available.
Continuity	Disaster recovery capacity.
Performance	Includes stress testing, peak analysis, capacity required and response times.
Recoverability	Business continuity requirements. Backup strategies and duplicated hw requirements.
Reliability/Safety	Assess if the system needs to be fail-safe. What happens if it fails?
Robustness	Ability to handle error and boundary conditions. For instance: network or hw issues, power outages, etc.
Scalability	Ability for the system to perform and operate as the number of users or requests increases (along time).

Structural Architectural characteristics

TERM	DEFINITION
Configurability	Ability for end-users to easily change aspects of the software configuration.
Extensibility	How important is to plug new pieces of functionality in.
Installability	Ease of system installation and it’s platforms.
Reuse	Ability to leverage common components across multiple products.
Localization	Support for multiple languages, units of measures or currencies.
Maintainability	How easy is it to apply changes to enhance the system.
Portability	Does the system need to run on more than one platform?
Supportability	What level of technical support is needed by the application?
Upgradeability	Ability to easily/quickly upgrade from a previous version to a newer version.

Cross-cutting Architectural characteristics

TERM	DEFINITION
Accessibility	Access to all your users, including those with any type of disability.
Archivability	Will the data be archived or deleted after a period?
Authentication	Security requirements to ensure users are who they say they are.
Authorization	Security requirements to ensure users can access only certain functions within the application.
Legal	What legislative constraints is the system operating in. What reservation rights does the company require? Regulations on building or deploying the application.
Privacy	Ability to hide transactions from internal company employees.
Security	Does the data need to be encrypted in the database? For network communications? Authentication for remote access?
Usability	Level of training required by users to achieve their goals with the application.

Comparison of Overlapping Characteristics

Characteristic	Focus	Key Question
Reliability	**Continuity**	How long can the system run without a failure?
Availability	**Readiness**	Is the system up and running right now when I need it?

Characteristic	Focus	Key Question
Interoperability	**Interaction**	Can two *different* systems exchange and use information?
Compatibility	**Coexistence**	Can two components perform their functions in the same environment?

Disclaimers

Any given list of characteristics will be incomplete.

Many of the terms are ambiguous and imprecise (for instance: interoperability and compatibility).

Many definitions overlap, such as availability and reliability.

The International Organization for Standardization (ISO) publishes a list of characteristics by capabilities.

Homework: Search the official ISO list and describe at least two that your system may have and why you consider them so.

Case study

Case study: Silicon sandwiches.

Description:

- A national sandwich shop wants to enable online ordering (in addition to its current call-in service).

Users:

- Thousands, perhaps one day, millions.

Requirements:

- Users will place their order, then be given a time to pick up and directions to the shop.
- If the shop offers delivery, dispatch the driver with the order to the user.
- Mobile-device accessibility.
- Offer national daily promotions/specials.
- Offer local daily promotions/specials.
- Accept payment online, in person or upon delivery.

Case study: Silicon sandwiches.

Additional context:

- Sandwich shops are franchised, each with a different owner.
- Parent company has near-future plans to expand overseas.
- Corporate goal is to hire inexpensive labor to maximize profit.

Silicon sandwiches.

Given the context an architect must think about the domain-level predictions about expected metrics, like the Users.

- This points out that we may want the application to be scalable and elastic.

Also, lets look at each of the requirements to see if they account for one or more specific characteristics:

- “Users will place their order, then be given a time to pick up and directions to the shop.”
- Here we need some third-party integration with a mapping app or service. This may impact reliability (the service may not be available), but is this really a characteristic concern?

Silicon sandwiches.

“If the shop offers delivery, dispatch the driver with the order to the user.”

- No special requirement here.

“Mobile-device accessibility.”

- This modifies the design of the application.
- But could account for mobile-optimization as well.
- Maybe consult with UX/UI and stakeholders to see if this is indeed a major concern.

“Offer national daily promotions/specials.”

“Offer local daily promotions/specials.”

- These both relate to customization since they need to change based on location.
- This customization can be either promoted to a characteristic or be tackled at the design level using a design pattern (like a Template Method).
- Again, everything is about the trade-offs.

Silicon sandwiches.

“Accept payment online, in person or upon delivery.”

- This implies security but, in this case, this is not a big concern.
- Security can be leveraged by using standard practices and reliable payment services.

“Sandwich shops are franchised, each with a different owner.”

- This suggests feasibility since not all the franchises are the same, for some the specific architecture may be too much.

“Parent company has near-future plans to expand overseas.”

- This leads to internationalization or i18n.
- Although this can also be handled at the design level.

“Corporate goal is to hire inexpensive labor to maximize profit.”

- This suggests usability, but, again, it refers more to the design.

What about performance?

- Should it be a characteristic?

Summary

We may need:

- Availability.
- Scalability.
- Elasticity.
- Customizability.
- Performance.
- Anything else?

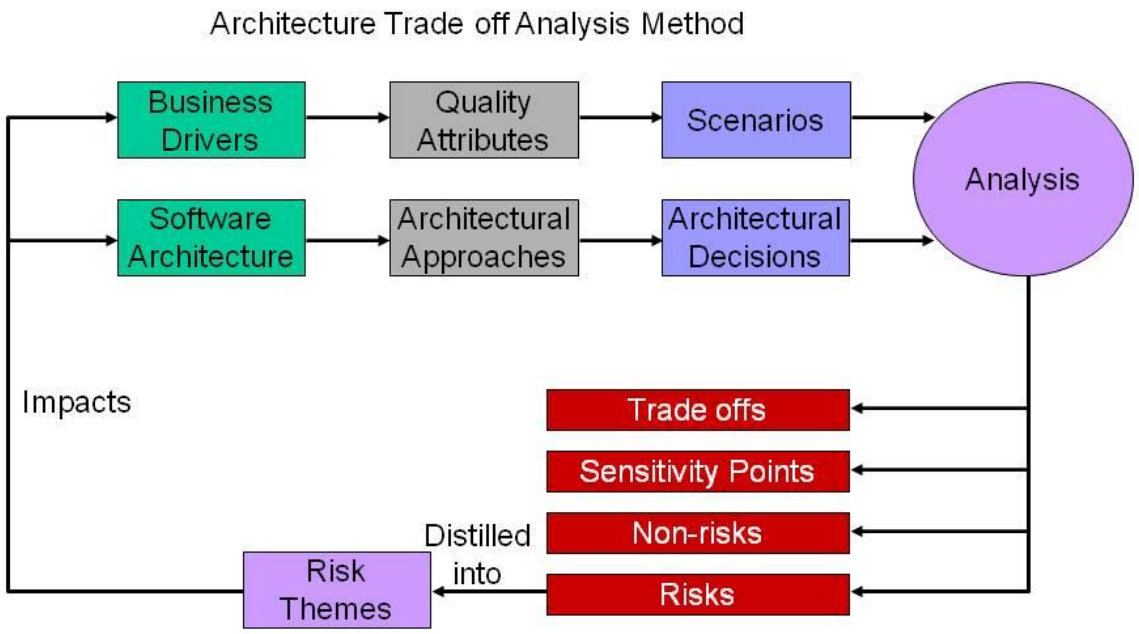
Quantifying and Prioritizing

ARCHITECTURE TRADE-OFF ANALYSYS METHOD

ATAM

ARCHITECTURE TRADE-OFF ANALYSIS METHOD

Architecture Trade-off Analysis Method



Steps & Outcomes

STEPS

Present the ATAM.

Present business drivers.

Present architecture.

Identify architectural approaches.

Generate quality attribute utility tree.

Analyze architectural approaches.

Brainstorm and prioritize scenarios.

Analyze architectural approaches.

Present results.

OUTCOMES

A concise presentation of the architecture.

Articulation of business goals.

The quality requirement in terms of a collection of scenarios.

Mapping of architectural decisions to quality requirements.

A set of identified sensitivity and tradeoff points.

A set of risks and non-risks.

A set of risk themes.

Quality Attribute Scenarios

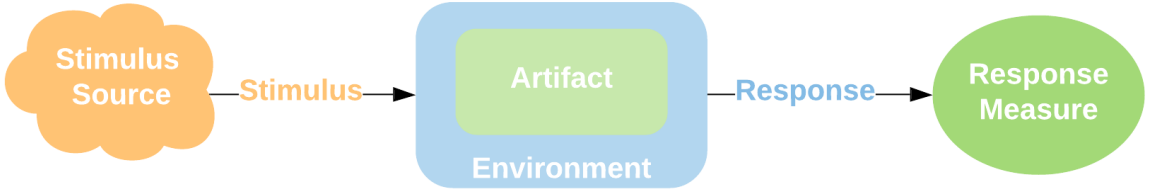
Why Scenarios?

Problem: "The system must be fast" is subjective and untestable.

Solution: Scenarios provide measurable, objective benchmarks

The 6-Part Anatomy

Part	Description	Example (Availability Scenario)
1. Source	Who or what generated the event?	An external monitoring service.
2. Stimulus	What is the actual event/condition?	The Primary Database instance crashes.
3. Artifact	What part of the system is affected?	The Persistence Layer / Database Cluster.
4. Environment	What is the system's current state?	During peak traffic hours (Normal Ops).
5. Response	How should the system react?	System switches to a "Read-Only" replica.
6. Measure	The measurable success criteria.	Switchover completes in **< 30 seconds**.



Silicon Sandwiches

Quality Attribute	Driving Business Requirement / Statement	1. Source	2. Stimulus	3. Artifact	4. Environment	5. Response	6. Measure
Scalability	"Users: Thousands, perhaps one day, millions."	Thousands of customers	Concurrent order volume increases to 1,000,000 users	Infrastructure / Order Processing System	Peak promotion period	System auto-scales resources to handle the throughput	All transactions processed without dropping orders
Availability	"User satisfaction" concern and the transition to "online ordering" as a primary service.	External monitoring service	Primary regional database instance crashes	Ordering Platform / Persistence Layer	Normal operations	System fails over to a redundant instance	Failover completes and service is restored in < 1 minute
Localization (i18n)	"Parent company has near-future plans to expand overseas."	Business expansion team	System is deployed to a new overseas market	User Interface & Database	Initial launch phase	System displays local languages, currencies, and units	100% of customer-facing text and currency symbols are localized

Silicon Sandwiches

Quality Attribute	Driving Business Requirement / Statement	1. Source	2. Stimulus	3. Artifact	4. Environment	5. Response	6. Measure
Usability	"Corporate goal is to hire inexpensive labor to maximize profit."	New, "inexpensive" store employee	Attempting to process a local promotion order	Point of Sale (POS) interface	High-volume "lunch rush"	Employee completes the transaction without supervisor intervention	Training time required is < 30 minutes; error rate is < 2%
Performance	"Users will place their order, then be given a time to pick up and directions."	Mobile user	Submits a completed sandwich order	Mobile Ordering API	Variable network conditions	System processes order and provides pick-up directions	Order confirmation and directions are displayed in < 2 seconds
Security	"Accept payment online" and "Sandwich shops are franchised, each with a different owner."	Malicious actor	Attempted unauthorized access to franchise owner records	Management Dashboard / Data Layer	Normal operations	System verifies identity and denies unauthorized access	0 unauthorized access incidents; all events are logged

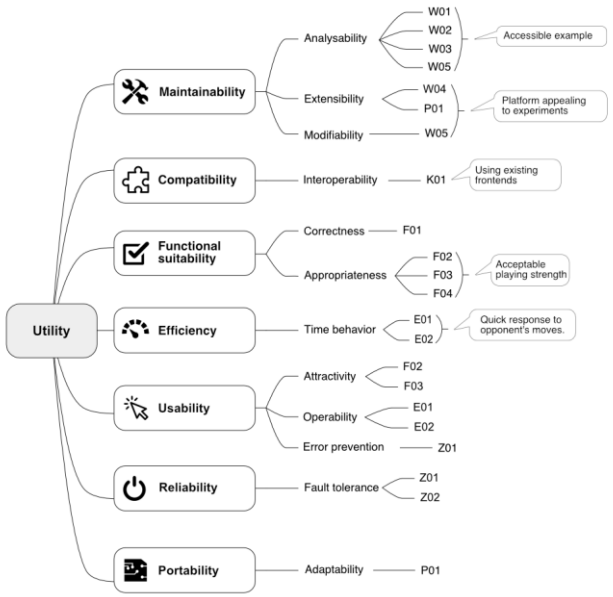
Prioritization & Utility Trees

What is a Utility Tree?

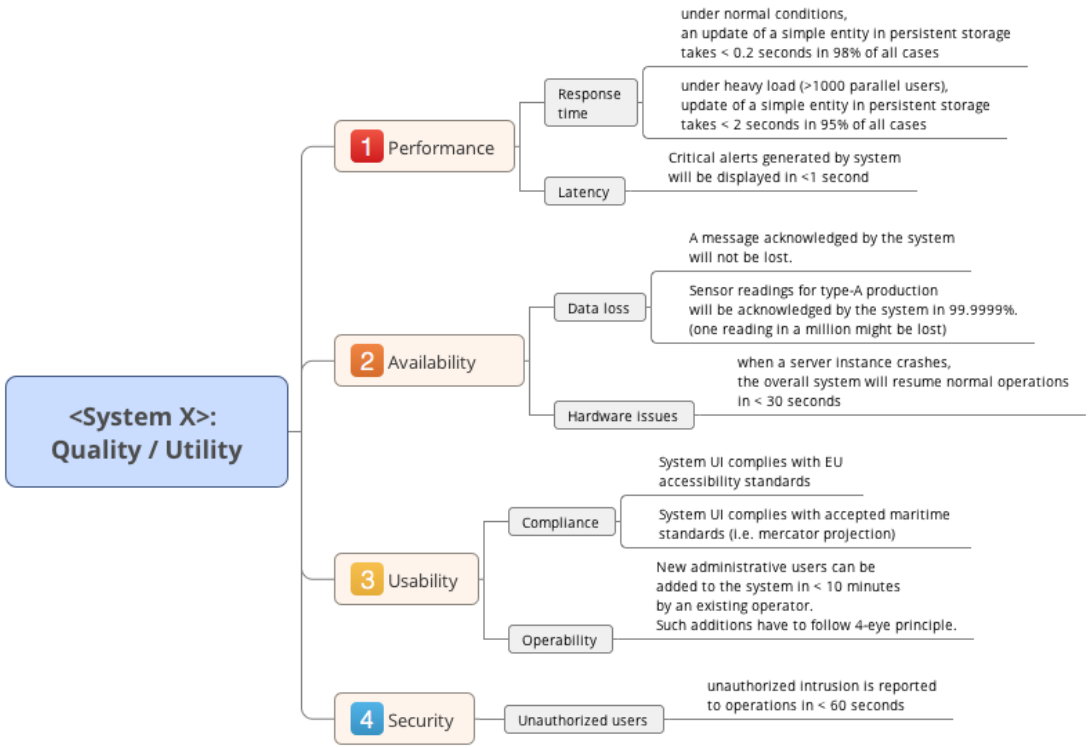
Definition: A top-down tool to translate broad business "Utility" into specific, measurable scenarios.

Structure:

- **Root:** Utility (The overall value of the system).
- **Branches:**
  - **Quality Attributes:** (e.g., Performance, Modifiability).
  - **Refinement:** (e.g., Data Latency, Ease of adding new payment methods).
  - **Scenario:** The 6-part scenario we built in Deck 5.



Utility tree example



Refinement and scenarios

Quality Attributes: (e.g., Performance, Modifiability).

Refinement: (e.g., Data Latency, Ease of adding new payment methods).

Scenario: The 6-part scenario we built in Deck 5.

Quality Attribute	Refinement	Scenario (The Leaf)
Availability	Server Failure	Database fails; system stays in "Read Only" mode (< 1 min).
Performance	Transaction Speed	Checkout completes in < 2 seconds under 5k concurrent users.
Modifiability	New Features	Adding a new shipping provider takes < 10 person-days.

Utility Tree (Silicon Sandwiches)

Quality Attribute	Refinement	Scenario (The Leaf)
Scalability	High Throughput	Concurrent order volume increases to 1,000,000 users during a national promotion; system auto-scales and processes all transactions without dropping orders.
Availability	Fault Tolerance	Primary regional database instance crashes; system switches to a "Read-Only" replica and restores service in < 1 minute.
Localization	Internationalization (i18n)	System is deployed to a new overseas market; 100% of customer-facing text, currencies, and units are correctly localized for the new region.
Usability	Learnability	A new, inexpensive employee attempts to process a local promotion order during a rush; they complete it with < 30 minutes of training and an error rate < 2%.
Performance	Transaction Latency	A mobile user submits a sandwich order; order confirmation and pickup directions are displayed in < 2 seconds.
Security	Authorization	A malicious actor attempts to access private records of a specific franchise owner; system denies access and logs the event with zero unauthorized data leaks.

The ASR Prioritization Matrix

Concept: Every leaf of the Utility Tree must be ranked.

The Two Dimensions:

- Business Value: How important is this to the stakeholder?
- Architectural Risk/Impact: How hard is this to implement or change later?

	Low Architectural Impact	High Architectural Impact
High Business Value	**"Quick Wins" (Prioritize Second)**	**"Architectural Drivers" (Top Priority)**
	Example: New User Profile Page UI	Example: Real-time fraud detection engine
Low Business Value	**"Fillers/Nice to Have"**	**"Technical Risks" (Prioritize Third)**
	Example: Color theme switcher	Example: Supporting legacy 32-bit OS

ASR Priorization Matrix (Silicon Sandwiches)

Scenario (Architectural Characteristic)	Business Value	Architectural Impact	Priority	Rationale
Scalability: Handle 1,000,000 concurrent users during promotions.	**High**	**High**	**1**	**Architectural Driver.** Defines the core infrastructure, database sharding, and caching strategy.
Availability: System failover to "Read-Only" mode in < 1 minute.	**High**	**High**	**1**	**Architectural Driver.** Crucial for "User Satisfaction". Requires complex redundant servers and data replication.
Performance: Mobile order confirmation and directions in < 2 seconds.	**High**	**High**	**1**	**Architectural Driver.** Significant for "Mobile-device accessibility". Often conflicts with security/encryption.

ASR Priorization Matrix (Silicon Sandwiches)

Scenario (Architectural Characteristic)	Business Value	Architectural Impact	Priority	Rationale
Security: Ensure franchise owners only access their own data.	High	Medium	2	**High Value.** Essential due to the franchised business model. Requires a robust authorization structure.
Usability: Inexpensive labor can use the POS with < 30 mins training.	High	Low/Med	2	**Quick Win.** A primary corporate goal to "maximize profit". While critical, it is often addressed at the design/UI level.
Localization: 100% localization for future overseas expansion.	Medium	Medium	3	**Technical Risk.** It is a "near-future" goal. Can be handled at the design level but adds complexity to the data layer.

Trade-offs

THE REALITY

The Architect's Dilemma

The Concept: You cannot optimize for everything. An architect is a manager of Scarcity (Time, Budget, Performance).Bullets:

The Conflict:

- High Security often reduces Performance (encryption overhead).
- High Availability increases Cost (redundant servers).

The Solution: Prioritization based on Architecturally Significant Requirements (ASRs).

The Architecture Trade-off Matrix

Goal	Usually Conflicts With...	Why?
High Security	**Performance**	Encryption and multi-factor checks add latency and processing overhead.
High Availability	**Cost / Complexity**	Requires redundant servers, data replication, and failover mechanisms.
High Flexibility	**Maintainability**	"Generic" solutions (as you noted) lead to complex code that is hard to debug.
High Performance	**Portability**	Optimization often requires hardware-specific code that won't run everywhere.

Trade-offs and least bad architecture

Applications can only support a few of the mentioned characteristics.

- Each supported one requires design effort and structural support.
- Each characteristic impacts others; for instance, security and performance.

Too many characteristics lead to generic solutions that rarely work because design becomes unwieldy.

Architects should strive to design architecture to be as iterative as possible.

Agile software development is about iteration.

"Never shoot for the best architecture, but rather the least bad architecture."

Trade-off Identification

Concept: Once you have your ASRs, you find the “Tension Points”.

Example Case:

- **ASR 1:** "Extreme Security" (Encryption at rest and in transit).
- **ASR 2:** "Sub-millisecond Latency.“
- **The Trade-off:** The architect must document that ASR 1 will negatively impact ASR 2, and the stakeholder must sign off on which one "wins."

Trade-off analysis (Silicon Sandwiches)

Tension Point	Goal A (High Priority)	Goal B (Conflicting Factor)	Why? (The Architectural Trade-off)
Availability vs. Cost	**High Availability** (99.9% uptime for online ordering)	**Cost / Complexity** (Franchise owner budget constraints)	Achieving high availability requires **redundant servers, data replication, and failover mechanisms**, which increases both the financial cost and the complexity of the system.
Performance vs. Security	**High Performance** (Sub-2s latency for mobile users)	**Security** (Encryption for online payments)	**Encryption and multi-factor checks** required for secure online payments add latency and processing overhead, which can slow down the user experience.
Scalability vs. Simplicity	**Scalability** (Handling millions of users)	**Simplicity / Time to Market** (Black Friday launch or budget)	Designing for extreme scale (throughput) requires complex infrastructure like **database sharding and caching**, which conflicts with the need for a simple, fast-to-deploy design.

Trade-off analysis (Silicon Sandwiches)

Tension Point	Goal A (High Priority)	Goal B (Conflicting Factor)	Why? (The Architectural Trade-off)
Customizability vs. Maintainability	**Customizability** (National vs. Local promotions)	**Maintainability** (Ease of fixing bugs/adding features)	Creating **"generic" solutions** to handle various local promotion styles often leads to complex code that is difficult for developers to debug or fix later.
Localization vs. Design Effort	**Localization / i18n** (Future overseas expansion)	**Structural Simplicity** (Least bad architecture)	Every additional characteristic supported **complicates the design**. Implementing internationalization early can make the initial system more unwieldy than necessary for a domestic launch.
Usability vs. Training Costs	**High Usability** (For "inexpensive labor")	**Design Resources** (Budget and Time)	High usability is defined by a **low level of training required**. Achieving this level of intuition in a complex POS system requires significant design effort and structural support

Conclusions.

There is not a right answer for the set of characteristics.

We must see which ones are key to success or must be enforced due to domain-concerns.

A useful exercise is to make another pass to the list of characteristics/requirements to see if we missed something.

ADRs

ARCHITECTURE DECISION RECORDS

The "Why did we do this?" Problem

The Scenario: You join a project. The database is a graph database (Neo4j), but the data is just simple user profiles. You ask "Why?". The only answer is "Dave thought it was cool in 2022." Dave left the company in 2023.

The Cost: Technical debt, confusion, and fear of changing anything.

The Solution: Architecture Decision Records (ADRs).

What is an ADR?

A short text file (usually Markdown) that captures a significant architectural decision.

It's **immutable history**: You don't edit an old ADR; you create a new one that supersedes it.

Stored in Git: Right next to the code it describes.

When do you write one?

Not for every function or variable name.

The "Significant" Rule: If the decision is hard to change later, affects multiple teams, or involves a trade-off between Quality Attributes (e.g., Performance vs. Cost).

The Anatomy of a Decision

Title: Short and descriptive.

Status: Proposed, Accepted, Superseded, Deprecated.

Context: What was the problem? What were the constraints? (The "Why").

Decision: What are we doing? (The "What").

Consequences: What is the "price" we pay? (The "Trade-off").

Benefits for the Team

Onboarding: New hires read the ADR folder to understand the system's "soul."

Alignment: No more circular arguments in meetings. "We already decided this in ADR-005."

Accountability: Ownership of the choice and its fallout.

ADR Template

ADR-00X: [Short, Descriptive Title]

Date: YYYY-MM-DD

Status: [Proposed | Accepted | Superseded by ADR-00Y]

Consulted: [Team Members / Stakeholders involved]

1. Context

Describe the technical/business problem we are facing. Include constraints (budget, time, existing tech stack). *Example: "We need to store 100k messages per second. Our current relational DB is locking up."*

2. Decision

State the choice clearly. Use active voice. *Example: "We will migrate the messaging module to Apache Cassandra."*

3. Alternatives Considered

Briefly list what else was looked at and why it wasn't chosen.

- *Redis (Too volatile for long-term storage)*
- *PostgreSQL Sharding (Too much operational overhead)*

4. Consequences

Positive (The Gains): What gets better? (e.g., High availability, horizontal scaling).

Negative (The Pains): What do we lose? (e.g., No more ACID transactions, team needs to learn CQL).